Fiscal Year:	FY 2009	**Task Last Updated:**	FY 06/30/2009
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Smolder Transition and Flaming (STaF) and Smolder Evaluation Test (SET)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	08/15/2005	**End Date:**	11/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Task added to Task Book in March 2009 when became aware of project per PI info		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05AA49A		
Performance Goal No.:			
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Task Description:	Smolder Transition and Flaming (STaF) and Smolder Evaluation Test (SET) The purpose of the program is to study smoldering and its transition to flaming in microgravity conditions. The experiment consists of a piece of the test fuel inserted in the wall of a flow duct. The back and sides of the sample are sealed and thermally insulated. The upstream end of the sample is heated by an igniter to initiate smolder propagation. The experimental measurements include the forward smolder propagation velocity, the smolder reaction temperature, and the permeability of the sample. The experimental parameters include the duct flow velocity, oxygen concentration, and external heat flux. The fuels to be studied include polyurethane foam, Pyrell stowage foam and other spacecraft materials.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This study investigated the effects of fire retardants, external heat flux, oxidizer flow velocity, and oxygen concentration on both the spontaneous and piloted transitions from smoldering to flaming in polyurethane foam. Smoldering produces flammable gases such as carbon monoxide and hydrocarbons that can be ignited through a spontaneous auto-ignition process (spontaneous transition) or by a pilot (piloted transition). Both modes of the transition from smoldering to flaming represent realistic and dangerous fire-ignition scenarios responsible for many residential and industrial fires. Tests were conducted on a non-fire retarded polyurethane foam and the fire retarded polyurethane foam Pyrell®. The experiments were conducted with small rectangular block samples oriented vertically in an upward wind tunnel. Three of the vertical sample sides were insulated and the fourth was exposed to an upward oxidizer flow and a radiant heat flux. For the piloted transition to flaming tests, a coiled resistance heating wire was used to ignite the smolder product gases. For both foams and both modes of transition to flaming, it was found that increasing the oxygen concentration, increasing the radiant heat flux, and/or decreasing the duct flow velocity increases the likelihood of a transition to flaming. Limiting external conditions for both modes of transition to flaming are reported for both foams. The results for both foams show that the piloted transition from smoldering to flaming is possible at lower oxygen concentrations than the spontaneous transition from smoldering to flaming. Thermocouple data, video imaging, infrared imaging, high-speed imaging, and schlieren imaging show that the spontaneous transition to flaming occurs in pores in the char region behind the smolder front in both foams. For the spontaneous transition to flaming in non-fire retarded foam, an ultrasound probing technique is used to measure the line-of-sight average permeability at various heights in the sample, in order to monitor the reactions occurring in the sample interior. It is found that the spontaneous transition to flaming is preceded by rapid fluctuations in permeability in the char region below the smolder front, indicating the formation of pores by secondary char oxidation. The secondary char oxidation in non-fire retarded foam is shown to provide the gaseous fuel, heat, and sites (large pores) for the gas-phase ignition. It is found that the time derivative of permeability, which provides a measure of the pore formation rate, indicates the likelihood of the spontaneous transition to flaming. The permeability data offers a means to track the propagation of the secondary char oxidation wave, and to calculate the secondary char oxidation velocity, which is found to be strongly correlated to the spontaneous transition to flaming. A simplified energy balance model is able to predict the dependence of the secondary char oxidation velocity on oxygen concentration and radiant heat flux. The experiments on the fire retarded foam Pyrell show that it undergoes a weak smoldering reaction that requires significant assistance in the form of external heat input. Gas chromatography results show that smoldering in Pyrell produces lower levels of CO and CO2 than smoldering in non-fire retarded foam, likely due to the fire retardant effects. Although the Pyrell was assisted with five times more igniter power than the nonfire retarded foam, the minimum oxygen concentration for the spontaneous transition to flaming was higher than that for the non-fire retarded foam. The results indicate that the fire retardants act in the solid phase to hinder the smolder reaction, and in the gas phase to impede the gas-phase ignition, however smoldering and the transition to flaming can occur in Pyrell at elevated oxygen concentrations. The experiments on the piloted transition to flaming in non-fire retarded foam show that the smolder velocity and peak smolder temperature are strongly correlated to the piloted transition to flaming. The minimum smolder velocity and temperature for the piloted transition to flaming are determined. The existence of a minimum smolder velocity for the transition to flaming supports the concept of a gaseous mixture reaching a lean flammability limit as the criterion for the transition, because the pyrolysate mass production rate is proportional to the smolder velocity. The critical mass production rate of pyrolysate is calculated from the minimum smolder velocity. The experiments on the piloted transition to flaming in Pyrell also indicate that the ignition occurs when the gaseous mixture at the pilot passes the lean flammability limit. Although the Pyrell was assisted with five times more igniter power and 30% more pilot power than the non-fire retarded foam, the minimum oxygen concentration for the piloted transition to flaming was significantly higher for both Pyrell foams than for the non-fire retarded foam. The results show that the fire retardants in Pyrell are very effective in preventing the piloted transition to flaming in normal air, but that Pyrell is susceptible to smoldering and the piloted transition to flaming in oxygen concentrations as low as 23%. These findings indicate that precautions should be taken in the design of applications of Pyrell in oxygen-enriched environments, such as those being considered from future NASA space missions, to reduce the risk of a transition to flaming. In addition to the experimental work, modeling efforts were performed to model smoldering combustion in two dimensions. Modeling smolder in two dimensions is the first step to being able to model the transition to flaming. Relatively few theoretical studies have examined the two-dimensional structure of the smolder wave. The two-dimensional smolder in polyurethane foam is modeled with a two-dimensional numerical formulation that includes a seven-step kinetic model of the polyurethane smolder reaction mechanism. The two-dimensional model formulation includes the effects of heat, mass, species, and momentum transfer of the porous solid and gas phase. The seven-step decomposition reaction mechanism, which includes a secondary char oxidation and an additional char pyrolysis step, was developed using optimization. The model was used to study the two-dimensionality of a forward propagating smolder wave. The model results show a two-dimensional structure in the temperature, species, and reaction profiles that agrees qualitatively with experimental observations. Oxygen is consumed at the reaction front, as expected, which leads to different reaction pathways governing the final products (i.e. thermal char and oxidative char). It was found that the model response is sensitive to boundary conditions, thermal properties, and heats of reaction for the char oxidation reaction. The incorporation of the secondary oxidation reaction step in the model paves the way to further analysis of the transition to flaming process.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Articles in Peer-reviewed Journals	Putzeys O, Fernandez-Pello AC, Rein G, Urban DL. "The piloted transition to flaming in smoldering fire retarded and non-fire retarded polyurethane foam." Fire and Materials 2008 Dec;32(8):485–99. http://dx.doi.org/10.1002/fam.981 , Dec-2008

Articles in Peer-reviewed Journals	Dodd AB, Lautenberger C, Fernandez-Pello AC. "Numerical examination of two-dimensional smolder structure in polyurethane foam." Proceedings of the Combustion Institute 2009;32(2):2497-504. http://dx.doi.org/10.1016/j.proci.2008.06.196 , Jan-2009
Articles in Peer-reviewed Journals	Rein G, Fernandez-Pello AC, Urban DL. "Computational model of forward and opposed smoldering combustion in microgravity." Proceedings of the Combustion Institute 2007 Jan;31(2):2677-84. http://dx.doi.org/10.1016/j.proci.2006.08.047 , Jan-2007
Articles in Peer-reviewed Journals	Putzeys O, Bar-Ilan A, Rein G, Fernandez-Pello AC, Urban DL. "The role of secondary char oxidation in the transition from smoldering to flaming." Proceedings of the Combustion Institute 2007 Jan;31(2):2669-76. http://dx.doi.org/10.1016/j.proci.2006.08.006 , Jan-2009
Articles in Peer-reviewed Journals	Rein G, Lautenberger C, Fernandez-Pello AC, Torero JL, Urban DL. "Application of genetic algorithms and hermogravimetry to determine the kinetics of polyurethane foam in smoldering combustion." Combustion and Flame, 2006 Jul;146(1-2):95-108. http://dx.doi.org/10.1016/j.combustflame.2006.04.013 , Jul-2006
Articles in Peer-reviewed Journals	Putzeys OM, Fernandez-Pello AC, Urban DL. "Ignition of combustion modified polyurethane foam." Journal of ASTM International 2006 Mar;3(3):Paper ID JAI13558. http://dx.doi.org/10.1520/JAI13558 , Mar-2006
Articles in Peer-reviewed Journals	Bar-Ilan A, Putzeys OM, Rein G, Fernandez-Pello AC, Urban DL. "Transition from forward smoldering to flaming in small polyurethane foam samples." Proceedings of the Combustion Institute, 2005 Jan;30(2):2295-302. http://dx.doi.org/10.1016/j.proci.2004.08.233 , Jan-2005
Articles in Peer-reviewed Journals	Rein G, Bar-Ilan A, Fernandez-Pello AC, Ellzey JL, Torero JL, Urban DL. "Modeling of one-dimensional smoldering of polyurethane in microgravity conditions." Proceedings of the Combustion Institute, 2005 Jan;30(2):2327-34. http://dx.doi.org/10.1016/j.proci.2004.08.150 , Jan-2005
Dissertations and Theses	Rein G. "Computational model of forward and opposed smoldering combustion with improved chemical kinetics." Ph.D. thesis, University of California, Berkeley, 2005. , Sep-2005
Dissertations and Theses	Putzeys O. "The spontaneous and piloted transitions from smoldering to flaming in polyurethane foam." Ph.D. thesis, University of California, Berkeley, 2007. , Jun-2007
Papers from Meeting Proceedings	Putzeys O, Fernandez-Pello AC, Urban DL. "Piloted ignition to flaming in smoldering fire-retarded polyurethane foam." 5th U.S. Combustion Meeting, San Diego, CA, March 2007. 5th U.S. Combustion Meeting, San Diego, CA, March 2007. , Mar-2007
Papers from Meeting Proceedings	Putzeys O, Titus R, Bar-Ilan A, Urban DL, Fernandez-Pello AC. "Observations of Forward Smoldering and the Transition to Flaming in Small Polyurethane Foam Samples with Ulrasound Probing." 43rd AIAA Aerospace Sciences Meeting , January 2005. Paper 2005-0715, 43rd AIAA Aerospace Sciences Meeting , January 2005. , Jan-2005
Papers from Meeting Proceedings	Rein G, Lautenberger C, Fernandez-Pello AC, Torero JL, Urban DL. "On the Derivation of Polyurethane Kinetics Parameters using Genetic Algorithms and its Application to Smoldering Combustion." 4th International Conference on Computational Heat and Mass Transfer, Paris (France), May 2005. Progress in Computational Heat and Mass Transfers, Paris, France, May 2005. Volume 2, p. 578-584. Proceedings of the 4th ICCHMT, Paris, Vol. 2, pp. 578-584, May 2005. , May-2005
Papers from Meeting Proceedings	Rein G, Lautenberger C, Fernandez-Pello AC, Torero JL, Urban DL. "Derivation of the Kinetics Parameters of Polyurethane Foam Using Genetic Algorithms." 4th Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia (USA), March 2005. 4th Joint Meeting of the U.S. Sections of the Combustion Institute, March 2005. Paper E02. , Mar-2005